

Natural Product *Zanthoxylum bungeanum* Based Multi-Functionalized Self-Polishing Interface for Sustainable Marine Antifouling

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Construction of an anti-biofouling coating on the surface has been proven to be an effective way to guarantee long-last anti-biofouling performance. However, amounts of heavy metal ions will be released, causing irreversible damage to the marine environment. In this work, a series of active components are extracted and isolated from *Zanthoxylum bungeanum* (*Z. bungeanum*), and their anti-biofouling characteristic and non-toxic to organisms are demonstrated with anti-barnacle larvae experiments and anti-algae tests. Furthermore, a novel resin contained with triclosan is prepared, and different self-polishing coatings are obtained by combining the pulverized materials of the *Z. bungeanum* fruit/seeds/leaves with the resin. The surface morphology, mechanical properties and anti-biofouling performance of the coatings are shown that incorporating the triclosan into self-polishing acrylic resin can improve broad-spectrum anti-biofouling properties, the algae adhesion densities (*Amphora sp.* and *Porphyridium sp.*) on the resin are almost to zero. According to the results of surface wettability, water resistance and anti-biofouling properties tests, it can be concluded that the coating with pulverized *Z. bungeanum* seeds (CWS) has a hydrophobic surface, it can continuously release antifoulant under seawater, exhibiting excellent anti-biofouling performances. This work therefore suggests *Z. bungeanum* can be used as a potential agent for developing environmentally friendly antifouling coatings.

platforms), leading to the biocorrosion^[1] and causing biofouling. The adhesion of marine organisms constructs biofilms that will further aggravate the severity of fouling and corrosion, increase the hull navigation resistance and the fuel consumption. In addition, the microorganisms adhering to the hull may also migrate with the transportations, causing biological invasion to the new ecological environment and disrupting the ecological balance.^[2] At present, anti-biofouling paints have been proved to be an effective method to protect them from the unwanted biofouling.^[3]

Tributyltin self-polishing copolymer paints (TBT-SPC paints), as the most effective anti-biofouling coating for inhibiting biofouling adhesion, can continuously release the contained toxic compounds into the seawater adjacent to the surface, any possible adhesive fouling would be hampered as a consequence. The International Maritime Organization (IMO) subsequently banned the use of TBT in 2008 in that TBT-SPC paints caused serious and irreversible pollution to the environment.^[4] In recent years, considering the substitution of toxic substances and environmental protection, great efforts from the coating design have been made to develop the low-toxic or even non-toxic bio-antifouling agents.

1. Introduction

Microorganisms in seawater often adhere to the surfaces of the marine equipment (e.g., ship hulls, oil well pipelines, drilling

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In nature, many organisms can secrete antimicrobial biopeptides to inhibit the adhesion of fouling organisms, including alkaloids, organic acids, terpenes, amides, etc. Although these components are also dispersed into the marine environment, they will not accumulate in the marine medium and cause ecological harm.^[5] To find environmentally friendly alternatives to the toxic anti-biofouling agents, researchers have isolated bioactive substances with anti-biofouling properties from nature or living organisms such as terrestrial and marine organisms.^[6] Rittschof and Tsukamoto extracted organic compounds such as lobster sarcosine, dichlorocarbimide sesquiterpenes, and guaiac sesquiterpenes from corals and sponges, respectively, which could effectively inhibit specific fouling organisms.^[7] However, it can only inhibit the adhesion of a few microorganisms, exhibiting poor broad-spectrum antibacterial activity. The compound produced by the marine fungus (*Aspergillus terreus* SCSGAF 0162) after solid-state fermentation had strong anti-barnacle adhesion activity, the comparative analysis of the structure and anti-biofouling activity of its analogs showed that the benzopyrone moiety may play a role in the anti-biofouling activity.^[8] In recent years, capsaicin extracted from capsicum has been reported as a natural non-toxic anti-biofouling biocide,^[9] research work has shown that capsaicinoids have more attractive properties than existing toxic anti-biofouling agents. However, the use and development of capsaicin anti-biofouling agent is restricted by many factors. On the one hand, there is a lack of capsicum varieties with high capsaicin content, the sources of raw materials are limited and expensive. On the other hand, the backward capsaicin extraction and processing technology makes it difficult to obtain high-quality capsaicin products. In general, the utilization rate of biological anti-biofouling agents is still relatively low.

It is also an effective method to further improve the interface resistance to biofouling under the premise of ensuring the degradation stability of the polymer resin by special functional modification of the self-polishing resin.^[2b,10] For one thing, the chemical structure of the self-polishing resin can be constructed to obtain a self-polishing anti-biofouling coating with different adsorption and activation energies, which can further regulate the self-polishing rate and anti-biofouling performance of the coating.^[11] For another, antibacterial agents can also be added to the self-polishing resin. The released antibacterial agent inhibits the activity of microorganisms in the nearby seawater, even kill microorganisms.^[12]

Triclosan is a synthetic bisphenol widely used in daily chemicals such as soaps, toothpaste, and other personal care

products. Triclosan could be used as antibacterial agents which effectively inhibited harmful bacteria: *Escherichia coli*, *Staphylococcus aureus*, *Candida albicans*, while protecting beneficial bacteria. At the same time, triclosan did not cause bacterial resistance. Although triclosan has been proved to have broad-spectrum antibacterial properties, and a large number of coatings containing triclosan have been studied in recent years,^[13] triclosan is currently identified as a pollutant in some countries.^[14] Therefore, in the actual use process, it is necessary to strictly control the amount of triclosan added.

Zanthoxylum bungeanum (*Z. bungeanum*) is a kind of edible plants, which has been used for culinary and medicinal purposes in China for centuries. The main chemical components of *Z. bungeanum* include volatile oils, alkaloids, amides, lignans, coumarins, flavonoids and polyphenols,^[15] several research have revealed that *Z. bungeanum* has antibacterial and antioxidant effects. In this study, we extracted a series of active components from *Z. bungeanum* and evaluated their anti-biofouling properties. Subsequently, a novel self-polishing resin containing triclosan was synthesized. Different self-polishing coatings were prepared by combining the pulverized materials of the *Z. bungeanum* fruit/seeds/leaves with the obtained self-polishing resin, the physicochemical properties and anti-biofouling properties of the coatings were evaluated. By combining the natural active ingredient with a broad-spectrum anti-biofouling painting, this study will provide a new design concept for fabricating environment-friendly self-polishing coatings.

2. Results and Discussion

2.1. Screening Active Components of *Z. bungeanum*

Due to its special morphological structure, life history and population ecology, barnacles have become one of the most important marine fouling organisms. They can generate a kind of adhesive called barnacle adhesive or barnacle cement, which is capable of adhering themselves tightly to different marine equipment.^[16] Therefore, we carried out a study on the effect of active components of *Z. bungeanum* (Figure S1, Supporting Information) on the attachment amount of barnacle larvae. The barnacle larvae were placed in the sterilized seawater containing different active components and cultivated for 120 h, after that, the larval attachment and death status of each group were statistically analyzed (Table 1).

Table 1. Adhesion and mortality of reticulated barnacle larvae.

Sample	Adhesion rate [%]	Death rate [%]	Sample	Adhesion rate [%]	Death rate [%]	Sample	Adhesive rate [%]	Death rate [%]
Blank ^{a)}	89.1	0	blank	84.9	0	blank	81.8	0
Control ^{b)}	88.9	0	control	85.5	0	control	80.9	0
CT-6 ^{c)}	57.4	0	TJ-1	89.7	0	HJZ-1	79.2	0
CT-12	89.2	0	TJ-2	56.2	0	HJZ-2	80.0	0

^{a)}Blank groups: 13 mL filter-sterilized seawater was used to cultivate reticulated barnacle larvae; ^{b)}Control groups: 1 mL mixture solvent (chloroform: methanol = 3:1(v/v)) was added into a 6 cm diameter Petri dish and made it evenly distributed on the bottom. After fully evaporation of the solvent, 13 mL filter-sterilized seawater was used to cultivate reticulated barnacle larvae; ^{c)}Experimental groups: the quantified compounds were dissolved in the mixture solvent (chloroform: methanol = 3:1(v/v)) to uniformly cover the bottom of a 6 cm diameter petri dish. After the solvent had completely evaporated, the content of the compound coated on the bottom of the petri dish was 10 µg cm⁻². After that, 13 mL filter-sterilized seawater was used to cultivate reticulated barnacle larvae.

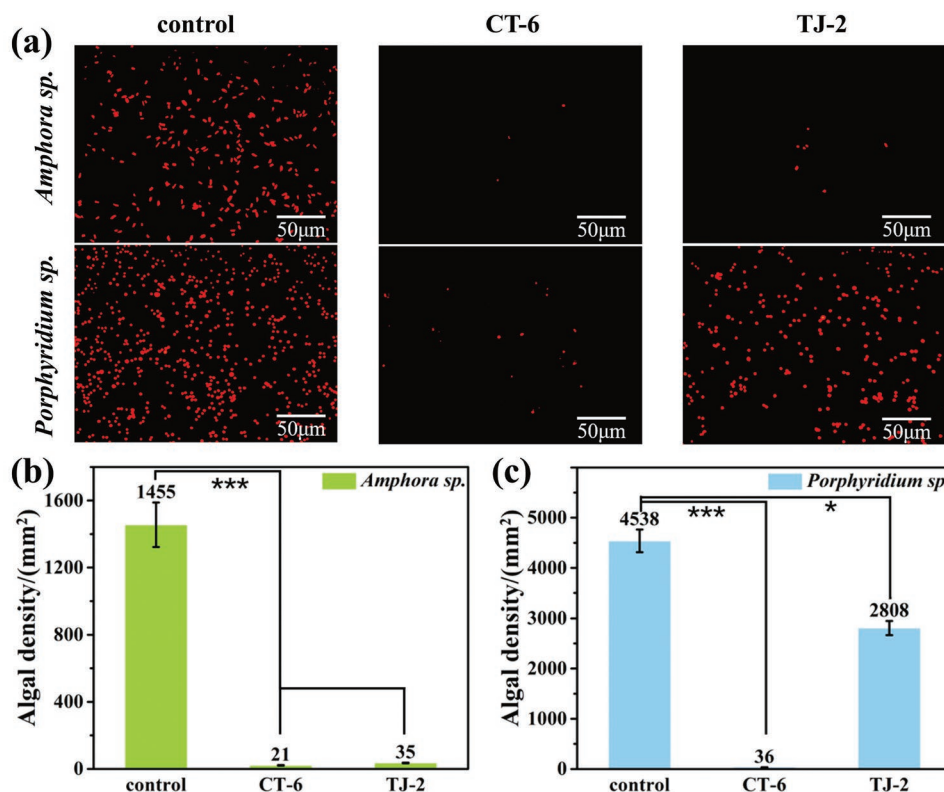


Figure 1. a) The fluorescence photos of *Amphora sp.* and *Porphyridium sp.* attachment on the glass slides and the general acrylic resin coatings embedded with the CT-6/TJ-2; statistic corresponding adhesive density of b) *Amphora sp.* and c) *Porphyridium sp.* (p values: $0.01 < *p < 0.05$, $0.001 < **p < 0.01$, $***p < 0.001$, all the values are expressed as mean $\pm$ SD, $n = 11$).

After three batches of the experiments, the results showed that there was no significant difference in the adhesion rate between the blank group and the control group in all experiments. While, CT-6 and TJ-2 could significantly inhibit the adhesion of the barnacle larvae, and the adhesion rate decreased from 89.1%, 84.9% to 57.4%, 56.2%, respectively. None of the barnacle larvae died after the experiments, indicating that CT-6 and TJ-2 were non-toxic. The rest of the components had no effect on larval adhesion, which not presented any anti-biofouling capability. Therefore, it was initially decided to select the extracted compounds CT-6 and TJ-2 for the following experimental characterization part.

To further evaluate the anti-biofouling properties of the active components of *Z. bungeanum*, we investigated the anti-biofouling performance of the general acrylic resin coatings embedded with the CT-6/TJ-2 against different typical fouling algae (*Amphora sp.* and *Porphyridium sp.*). It can be seen from Figure 1. that both of the two compounds had obvious inhibitory effect on *Amphora sp.*, making the adhesion rate less 1% than the blank sample (the adhesion amount decreased from 1455, 4538 to 21, 36 respectively). CT-6 also had good anti-adhesion effect to *Porphyridium sp.*, the anti-adhesion rate reached more than 97%. By contrast, TJ-2 exhibited poor anti-adhesion performance to *Porphyridium sp.*, and the amount of adhesion was about half of the blank sample, the anti-adhesion rate was only $\approx 38\%$. It can be concluded that CT-6 had better broad-spectrum anti-biofouling properties than TJ-2. Therefore, the compound CT-6 was further screened as the main anti-biofouling active agent for the next experiments.

QCM experiment was carried out to further evaluate the influence of CT-6 solution concentration on algal adhesion behaviors. The CT-6 solutions with different concentrations were prepared to cultivate *Amphora sp.* and *Porphyridium sp.* algae for 12 h, after that, the adsorption quality of the algal solutions on the quartz wafer was tested. The influences of CT-6 solution concentration on the adsorption behaviors of different typical fouling algae were shown in Figure 2 (the adsorption-desorption curves in Figure S2, Supporting Information). For *Amphora sp.* and *Porphyridium sp.*, both of the adsorption and desorption mass for CT-6 solution were lower than that of the control group, the net adsorption mass of algae decreased firstly then increased as the CT-6 solution concentration increased. When the CT-6 concentration was 1.5 mg mL^{-1} , the lowest net adsorption mass for *Amphora sp.* and *Porphyridium sp.* were 5.31 and 42.48 ng cm^{-2} , respectively. The obtained results were consistent with those shown in Figure 1, which further confirmed the antifouling properties of CT-6. In addition, we speculated that when the content of active components CT-6 was too high, the increase of net adsorption might be related to other factors, such as the dispersion characteristics and solubility of active components.

2.2. Synthesis of Self-Polishing ISO-Tri Resin

The above results proved that there were active components with marine anti-biofouling properties in *Z. bungeanum*.

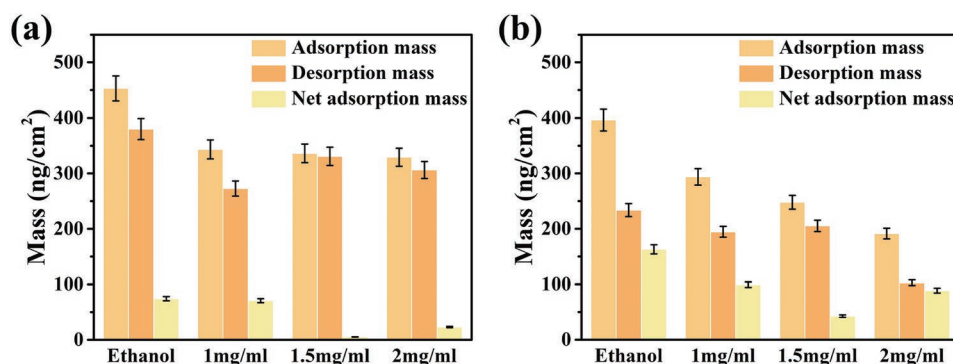


Figure 2. QCM-D adsorption mass of algae solutions a) *Amphora sp.*, b) *Porphyridium sp.* with different CT-6 concentrations.

the coatings fabricated by mixing CT-6 and acrylic resin also exhibited resistance to the adhesion of different algae. Next, for a more expected comprehensive anti-biofouling ability, we synthesized a novel self-polishing resin with triclosan chains. Meanwhile, considering the covalently conjugation of triclosan with acrylic resin, triclosan was modified with Isocyanatoethyl methacrylate (ISOMA) to introduce the polymerizable functional group, and the synthesis process of the modified triclosan (ISO-Tri) was shown in **Figure 3**.

The ^1H NMR and FTIR spectra of ISO-Tri were examined to explore any changes after the modification of the triclosan (in Figures S3 and S4, Supporting Information.). Compared with the unmodified triclosan, the absorption peaks of $=\text{CH}_2$ groups at 3080 cm^{-1} and the absorption peaks of $\text{C}=\text{C}$ groups at 1675 cm^{-1} were observed for ISO-Tri. Since the $-\text{OH}$ group in triclosan had been reacted, the adsorption peak of hydroxy group at 3324 cm^{-1} became narrow and sharp which was attributed to the $-\text{CONH}-$ adsorption. Furthermore, the adsorption peaks at 2917 cm^{-1} were assigned to the $-\text{CH}_2-$. The peaks on the ^1H NMR correspond to the H atoms of the ISO-Tri. ISO-Tri: ^1H NMR (400 MHz, CDCl_3) δ 7.38 (dd, $J = 2.5\text{ Hz}$, 1H), 7.19 (s, 1H), 7.17–7.04 (m, 2H), 6.80 (d, $J = 8.8\text{ Hz}$, 1H), 6.79–6.70 (m, 1H), 6.04 (s, 1H), 5.52 (t, $J = 1.8\text{ Hz}$, 1H), 5.24 (t, $J = 6.1\text{ Hz}$, 1H), 4.16 (t, $J = 5.3\text{ Hz}$, 2H), 3.44 (q, $J = 5.6\text{ Hz}$, 2H), 1.87 (s, 3H). The ^1H NMR and FTIR spectra of ISO-Tri confirmed the successful grafting of methacrylate groups onto triclosan.

2.3. Anti-Algae Adhesion Test of Self-Polishing ISO-Tri Resin

As shown in Figure 1, few algae were covered on the general acrylic resin with the active component, indicating that the resin embedded with the active components (CT-6 and TJ-2) had a cer-

tain anti-algae fouling performance. For the purpose of further improving its static anti-biofouling effect, we introduced triclosan molecules, a broad-spectrum antibacterial agent, into the self-polishing resin. The active components CT-6 and TJ-2 were respectively added to the ISO-Tri self-polishing resin and left at room temperature for 24 h to fully cure. In order to evaluate their anti-biofouling capabilities, different algae were employed, and the corresponding fouling behaviors were presented in **Figure 4**. Compared to the general acrylic resin coatings, under the synergistic action of triclosan and active components, the self-polishing resin with CT-6 could effectively reduce the algae fouling, the adsorption values of *Amphora sp.* and *Porphyridium sp.* on the ISO-Tri resin (CT-6) were basically 0, 0, respectively. Furthermore, ISO-Tri resin (TJ-2) had a lower density of *Porphyridium sp.* compared to the general resin coatings, confirming that the addition of triclosan could enhance the antifouling performance from another point of view.

2.4. Physicochemical Properties of the Prepared Self-Polishing Coatings

In the actual application process, it is necessary to add a large amount of active components to the resin. However, the extraction procedure is complicated and the yield of the extracts is relatively low. In order to address that problem, we tried to grind the capsicum and the fruits/seeds/leaves of *Z. bungeanum* into fine powders, then blended them with self-polishing resin to form different coatings (The obtained coatings were denoted as CWC, CWF, CWS, and CWF, respectively. In addition, coatings without the pulverized material of *Z. bungeanum* were the Control groups). The coating adhesion force (CAF) was carried out by pull-off tests, and the obtained experimental data were presented in **Figure 5**. All of the fracture surfaces were located

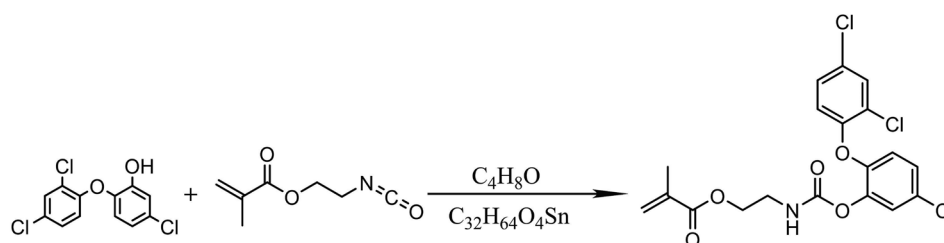


Figure 3. The synthesis process of ISO-Tri.

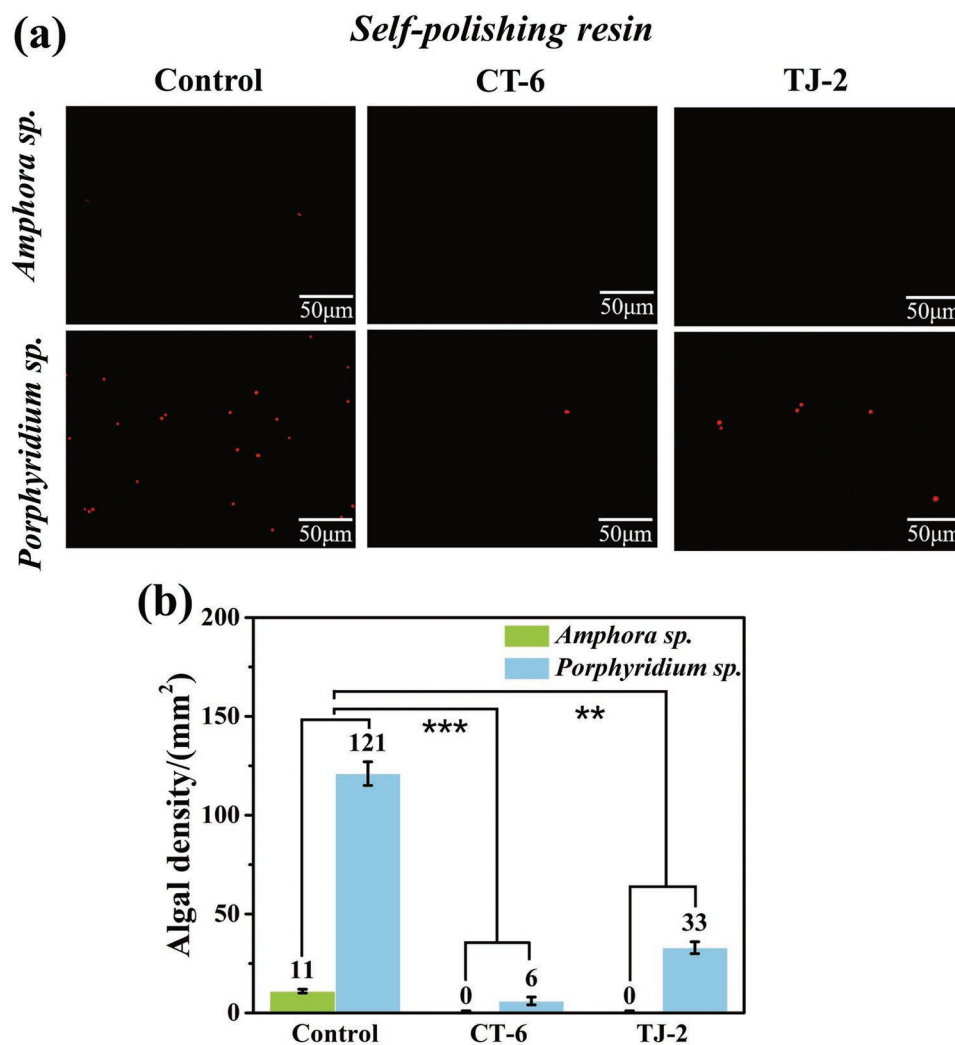


Figure 4. a) The fluorescence photos of *Amphora sp.* and *Porphyridium sp.* adhesive on the ISO-Tri self-polishing resin coating and the coatings with the CT-6/TJ-2; b) statistic corresponding adhesive density of *Amphora sp.* and *Porphyridium sp.* (p values: $0.01 < *p < 0.05$, $0.001 < **p < 0.01$, $***p < 0.001$, all the values are expressed as mean $\pm$ SD, $n = 11$).

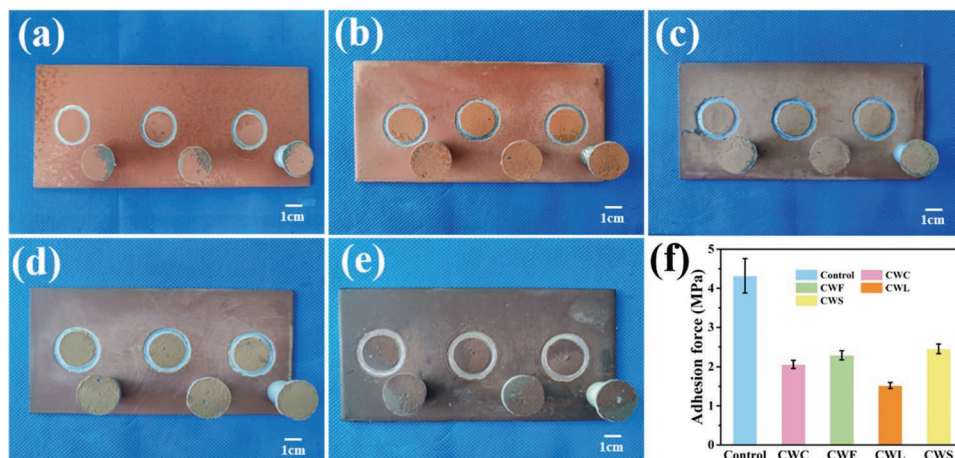


Figure 5. Coating adhesion measurement by pulled off tests a) Control, b) CWC, c) CWF, d) CWL and e) CWS; their corresponding average value (f).

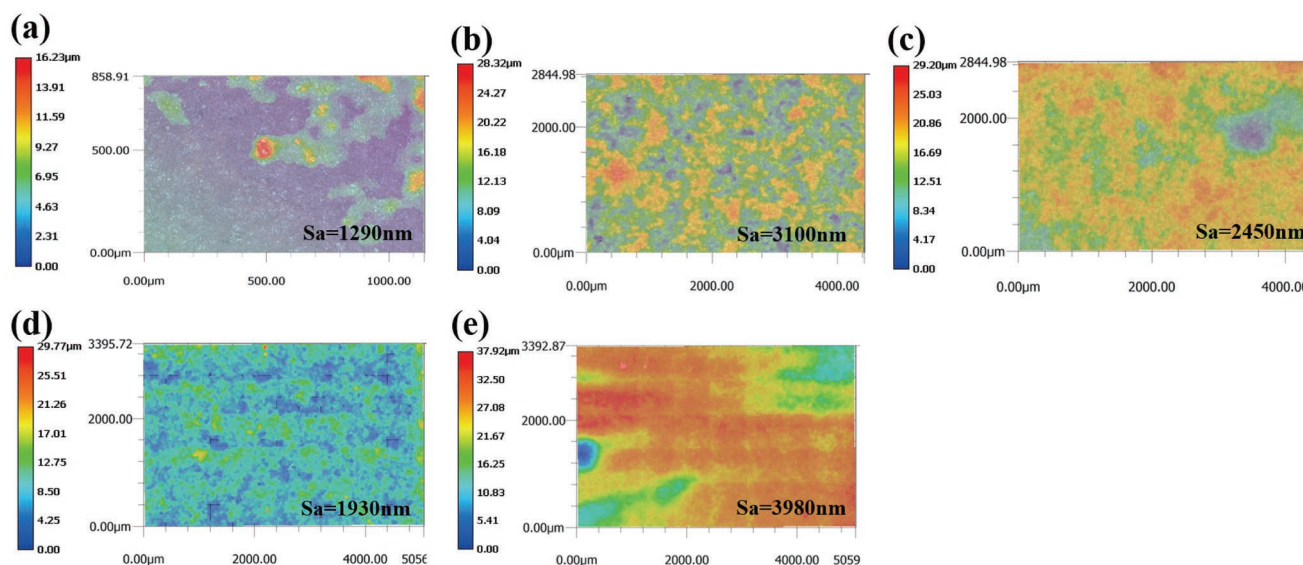


Figure 6. Surface roughness of the different coatings: a) Control b) CWC, c) CWF, d) CWL, and e) CWS.

above the tie-coatings, indicating that the anti-biofouling coatings could firmly bond to the substrates. After adding pulverized materials to the coatings, the values of CAF decreased. The CWL coatings had the lowest CAF ≈ 1.52 MPa, while the CAF of the CWS coating was 2.45 MPa. Although there was still a certain gap of the CAF between above coatings and commercial paints, they could also meet the needs of engineering applications.

Surface topographies of the five different coatings were shown in **Figure 6**. It could be seen that the surfaces of the coatings were flat and dense, and the microstructure was evenly distributed. The obtained roughness data indicated that all of the coatings mixed with different grounded materials had good film-forming properties, the CWS coating had a highest roughness (~ 3.980 μm) than the others.

It is well known that surface characteristics have important impacts on the antibiofouling performance of coatings, especially, the antibiofouling mechanisms of the coating are also subject to the surface wetting to some extent.^[17] Therefore, the interfacial wetting behaviors of the as-prepared coatings were further investigated. The corresponding contact angles (CAs) of polar water and non-polar diiodomethane were shown in **Figure 7**. From the point of view of CAs, the CWC and CWF were hydrophilic, which guaranteed more outstanding self-polishing and self-renewal capabilities of coating and further might endow the coating with preferred anti-biofouling ability. Nevertheless, considering the non-polar components contained in *Z. bungeanum* seeds and leaves, surfaces with hydrophobicity but lipophilic abilities for CWL and CWS were obtained. As shown in the **Figure 7**, the CAs of water on the CWL and CWS were larger than 90° . Moreover, the higher content of non-polar substances in the *Z. bungeanum* seeds was easier to accumulate on the surface and even form an oil film. Different from the other three samples, the CWS therefore made a higher water contact angle and smaller diiodomethane contact angle, and might ensure an excellent fouling release and water-proof performances.

As displayed in **Figure 8**, the adhesion rates of algae for the ISO-Tri coating were 0.76% (*Amphora sp.*) and 2.02% (*Porphyridium sp.*). After adding the pulverized materials into the resin, all of the four coatings showed excellent anti-adhesion properties to the different algae. The algal density (*Amphora sp.*) for CWC and CWL coatings was reduced to zero, by contrast, CWF and CWS coatings exhibited slightly higher values. In addition, the algal density (*Porphyridium sp.*) for CWL coatings was relatively lower than the others. The obtained results were highly consistent with those found in **Figure 4** because CT-6 was extracted from the *Z. bungeanum* leaves. According to these tests, it confirmed that the grounded materials of *Z. bungeanum* could be added as an anti-biofouling additive to the self-polishing coatings, which exhibited excellent anti-biofouling properties.

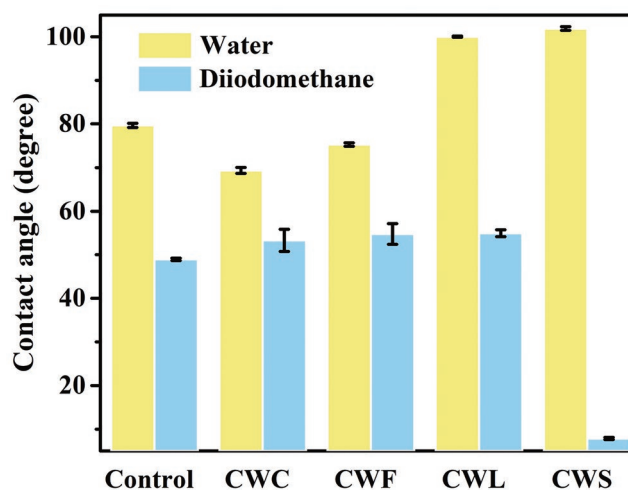


Figure 7. Contact angles of water and diiodomethane of the five coatings.

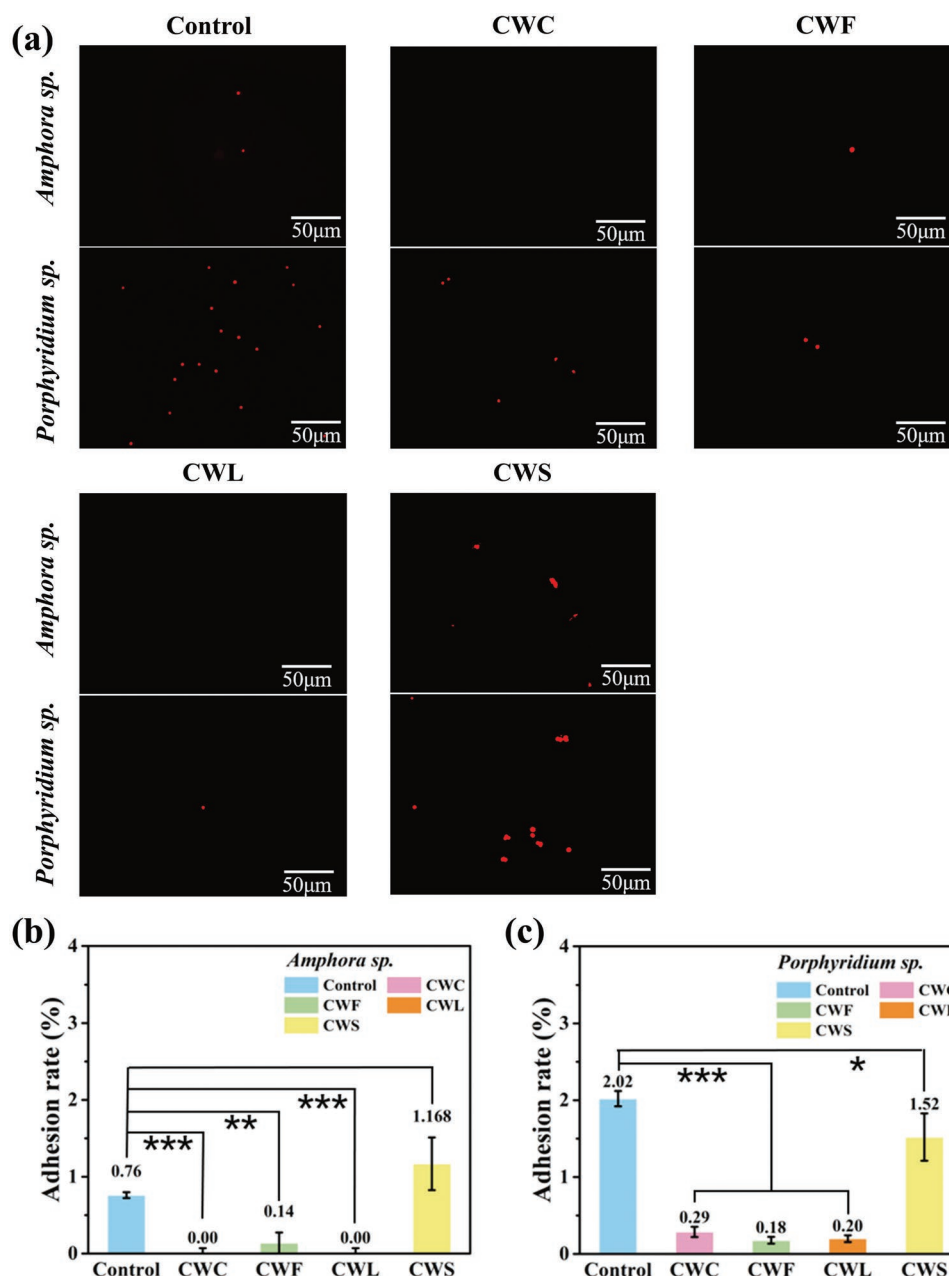


Figure 8. a) The fluorescence photo of *Amphora sp.* and *Porphyridium sp.* adhesion on the five coatings; the relative percent of microalgae b) *Amphora sp.* and c) *Porphyridium sp.* adhering to the surface when compared with glass sample. (p values: $0.01 < *p < 0.05$, $0.001 < **p < 0.01$, $***p < 0.001$, all the values are expressed as mean $\pm$ SD, $n = 11$)

2.5. Characterization of Water Resistance and Release Behavior of Active Substance

The water resistance ability was a key point to affect the stability of the coatings in a marine environment, the related experiment was carried out by immersing the coatings in the artificial seawater for four weeks. From the results of water immersion (Figure 9), it could be concluded that the water resistance of the coatings was related to the ground powders. The pulverized material of *Z. bungeanum* seeds was an oily substance, its hydrophobic property was the most important

factor in improving the water resistance of the CWS coatings, few defects could be found on the CWS coating after being immersed in water for a long time. However, the water resistance of the CWL coating was poor. The coating peeled off only after one week because the leaves would be swelled after water absorption, which broke the bond of the coating to the substrate. As for the CWF coating, humps and micro-cracks gradually appeared as the experiment proceeded.

Basing on the self-polishing character of the as-prepared anti-biofouling coatings, the controllable and stable self-refresh capability would be ensured, which therefore endowed the

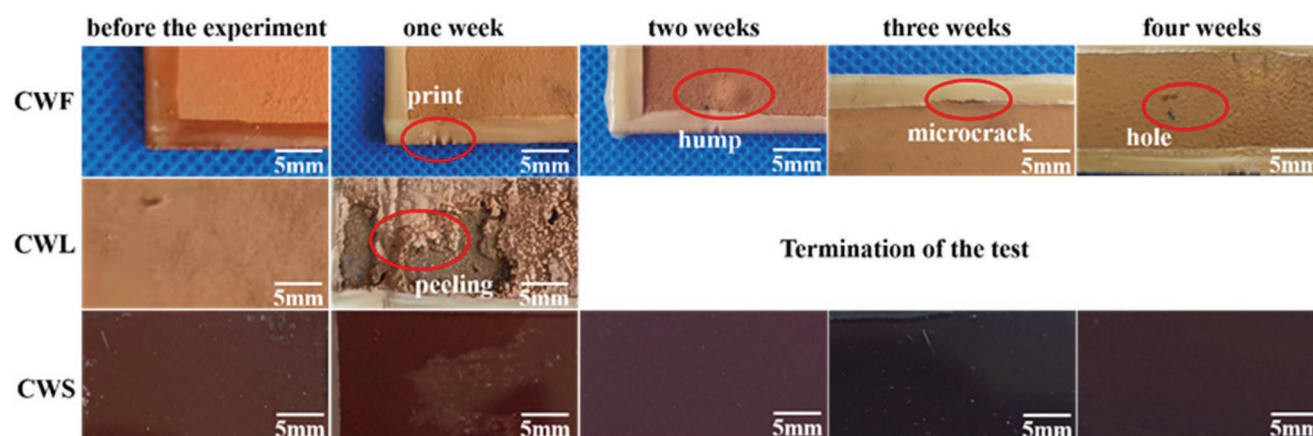


Figure 9. Surface topographies of the a) CWF, b) CWL, c) CWS coatings immersed in the artificial seawater.

coatings with sustainable release behaviors of active antifoulant *Z. bungeanum*. Here, in order to further prove the degradable characteristic of antifoulant *Z. bungeanum*., the accumulated concentrations of the active components during self-polishing process were investigated at different submerging times through UV-spectrophotometer, and the obtained experimental results were showed in **Figure 10**. Before the characterization, the standard adsorption curve for the active component *Z. bungeanum*. was determined as shown in Figure S5 (Supporting Information), by which the accurate concentration of accumulated active antifoulant could be defined quantitatively. As shown in the Figure 10, the release rate of active antifoulant for the CWF sample was faster than the CWS sample, which was mainly attributed to the preferred hydrophilicity of CWF sample. Besides, the accumulated concentration of antifoulant for the CWF decreased and then kept constant in the next few weeks ($\approx 0.011 \text{ mg mL}^{-1}$). Meanwhile, its corresponding feature UV adsorption peak gradually flattened, indicating an irreversible change of molecular structure of antifoulant *Z. bungeanum*. This experimental phenomenon confidently proved that the antifoulant had degraded with the submerging continuing. More interestingly, the release and degrading rate of active component for the CWF sample equaled each other which was confirmed by the constant *Z. bungeanum*. concentration after the two weeks immersion. However, for the CWS sample, it was obviously to find that degradation behavior of antifoulants was different from the CWF sample besides the release behavior. In detail, the accumulated concentration of antifoulant increased further in the following three weeks

(the corresponding highest concentration was 0.019 mg mL^{-1}) and then decreased. The corresponding UV feature adsorption peak became flatted slightly while compared with the CWF sample, indicating a slower degrading process of antifoulant *Z. bungeanum*. The quiet distinct degradation behaviors absolutely meant different degradation mechanisms for the two samples. As shown in the Figure S6 (Supporting Information), different from the CWS sample, the acquired solution became dark more and more for the CWF sample, which might confirm that there was extra factor accelerating the *Z. bungeanum*. degradation for the CWF coating, such as the colored groups component dissolved in the solution.^[17] As a consequence, the synthesized self-polishing anti-biofouling coatings not only could release the active antifoulant preventing the biofouling also the released *Z. bungeanum*. antifoulant could degrade quickly, these merits would guarantee the coating expected anti-biofouling property and avoid any possible environmental issue induced by the plenty use of antifoulants.

Based on above characterizations results, the possible anti-biofouling mechanism of *Z. bungeanum* based self-polishing coatings was proposed (**Figure 11**). As soon as the coating was immersed in seawater, macroproteins and marine microorganisms would adhere to its surface. Along with the coating fell off under the scouring action of seawater, the *Z. bungeanum*. antifoulant would diffuse out. These *Z. bungeanum* antifoulant not only inhibited the adhesion of marine microorganisms on the coating, but also protected the environment due to their non-toxic properties.

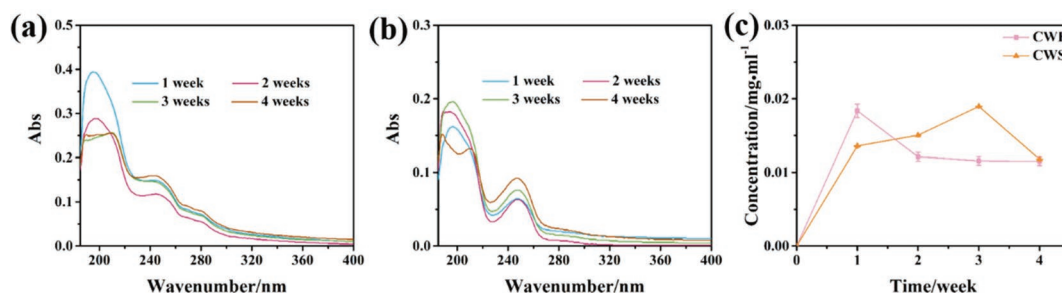


Figure 10. UV absorption curve of coating self-polishing solution: a) CWF, b) CWS; and c) the concentration of active substance in solution.

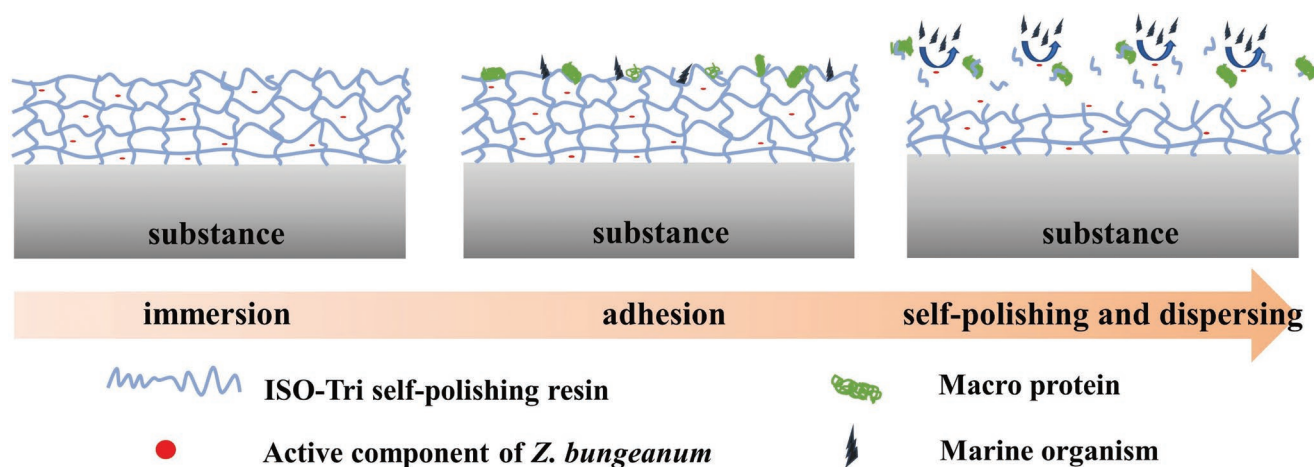


Figure 11. Suggested anti-biofouling mechanism of *Z. bungeanum* based self-polishing coatings.

3. Conclusion

In this study, functionalized triclosan with anti-algae effect was co-polymerized with general acrylic molecules to fabricate the self-polishing resin by free radical polymerization. Natural active antifoulant *Z. bungeanum* with self-degradation character was screened. The essential anti-adhesion test and QCM-D adsorption experiment proved that the CT-6 of *Z. bungeanum* had nontoxic and broad-spectrum anti-adhesion properties. Then, a multifunctional integrated anti-biofouling coating with good anti-biofouling effect and environmental friendliness was constructed by incorporating the active component with the self-refreshing resin. The obtained anti-biofouling coating could greatly remove the fouling algae adhered to the surface by the synergetic effect of self-polishing and sustainable release of the anti-biofouling active agent *Z. bungeanum*, which has the effect of either expelling marine microorganisms (both green algae and diatom) or further reducing the adhesion of algae on the surface of coating. Besides, surface properties test, mechanical performance test and coating water resistance test showed that the as-prepared coating had a homogeneous microstructure, tough mechanical properties, and good water resistance. Basing on the above experiments, it is believed that *Z. bungeanum* can be used as an effective anti-biofouling inhibitor in coatings, and CWS is the most suitable raw material for anti-biofouling coatings, which is a great inspiration for the next generation of environmentally friendly antifouling coatings.

4. Experimental Section

4.1. Experimental Materials

Zanthoxylum armatum DC seeds were collected from Sichuan province (Meishan), *Zanthoxylum bungeanum* Maxim. leaves and fruits were collected from Gansu province (Tianshui), all of them were dried before use. Triclosan (AR grade), ethanol (EtOH, AR grade), methyl methacrylate (MMA, AR grade), butyl methacrylate (BMA, AR grade) and hydroxyethyl methacrylate (HEMA, CP grade) were purchased from Macklin.

Dibutyltin dilaurate (DY-12, CP grade) and isobornyl methacrylate (IBOMA, CP grade) were purchased from Meryer. Tetrahydrofuran (THF, CP grade), 2-methylpropionitrile (AIBN, AR grade), and 1-dodecanethiol (AR grade) purchased from Energy Chemical. Isocynoethyl methacrylate (ISOMA, AR grade) was purchased from TCI (Shanghai). Ethyl Acrylate (EA, AR grade), butyl acrylate (BA, AR grade), and dimethylbenzene (DMB, AR grade) were purchased from XiLong Scientific. 2-Methoxyethanol (MOE, CP grade) was purchased from Offset Corporation. Kypzin, zinc oxide, ferric oxide, barium sulfate, and organic bentonite (OMMIT) were purchased from J&K Chemical Ltd. Natural rosin was purchased from Shanghai Sangjing Chemical Co. Ltd. BYK-163, BYK-104S, BYK-052, and BYK-354 was purchased from BYK Additives & Instruments. Artificial seawater was prepared by diluting natural sea salt in water. Different typical fouling algae (*Porphyridium sp.* and *Amphora sp.*) and their respective culture medium, Kock and Erdschreiber, were provided by the Freshwater Algae Culture Collection at the Institute of Hydrobiology, the algae were cultivated in a constant temperature and humidity incubator. All the chemicals were used without further purification.

4.2. Extract Active Components of *Z. bungeanum*

The active components extracted from *Z. armatum* DC fruit were named as TJ-1, TJ-2. The active components extracted from *Z. bungeanum* Maxim. leaves were named as HJZ-1, HJZ-2. CT-6 was the pure product isolated from leaves, CT-12 was a derivative of CT-6 synthesized through chemical methods.

One kilogram of dried *Zanthoxylum armatum* DC was immersed in 4.0 L EA at room temperature for 7 days and the process was repeated for three times. After the immersed extract operations, the obtained extract solution was further concentrated by a rotary evaporator at 40 °C for vacuum decompression until no EA was detected. The product was named TJ-1. The residue of above extract process was further extracted by water at 60 °C for three times, and the water extract was concentrated and dried. The water extract was named TJ-2 and stored.

Five kilogram of *Zanthoxylum bungeanum* Maxim. Leaves was dried in the shade area and pulverized to a size of 0.1 mm by a grinder. The powder of *Z. bungeanum* Maxim. leaves were extracted with water for three times at a 350 W, 60 °C, for 50 min in ultrasonic extracting device. The mass ratio of *Z. bungeanum* Maxim. leaves and water were 1:50. The extract of the mixture was filtered, and collected the supernatant. Further, the obtained extracts solution was concentrated and dried by a rotary evaporator at 40 °C for vacuum decompression, and the product was named A. The crude extract(A) was fully dissolved in water and separated through a D101 resin. Briefly, the water solution of crude extracts was added to the column to adsorption for 24 h by a standing, and then D101 adsorbent resin was respectively eluted with three times BV of pure water, 30% EA, 60% EA, 80% EA, and EA. The eluants were collected and concentrated as the above-mentioned operation, and each residue was numbered HJZ-1 (pure water fraction), HJZ-2 (30% EA fraction), CT-X (80% EA fraction).

The fraction CT-X (80% EA) was dissolved in a small amount of MA, and then, it was mixed with silica gel. The wet column loading method and dry sample loading method were selected. Gradient elution of dichloromethane and acetone (30:1-5:1) was used. Sample solutions were received in equal quantities, and similar components were detected and combined with thin layer chromatography. Finally, removing the solvent a yellow oil product named CT-6 was obtained. Another, gradient elution at the chloroform-ethyl acetate system (10:1, R_f = 0.4) was used for a silica gel column elution. The same fractions were collected according to the results of thin layer chromatography and the obtained product was named CT-12.

4.3. Synthesis of General Acrylic Resin Embedded with Active Components

5 g DMB and 1 g BA was added to three-necked bottom, and the temperature was kept at 80 °C. Dipped the monomer solution (9 g MMA, 3 g HEMA, 2 g styrene, 5 g BA, 0.4 g AIBN,

and 15 g DMB) within 2 h and kept for 2 h. The solution (0.05 g AIBN was dissolved 2 g DMB) was added into three-necked bottom and reacted 1 h, and the acrylate resin were obtained. Finally, CT-6/TJ-2 were mixed into the acrylate resin and made the mass fraction of CT-6/TJ-2 in the resin reach 10 wt. %.

4.4. Synthesis of the Modified Triclosan (ISO-Tri)

14.50 g (0.05 mol) triclosan was dissolved in 60 mL THF and added 0.20 mL DY-12, the mixture solution was stirred at 55 °C. Subsequently, 7.75 g (0.05 mol) ISOMA was dissolved in 20 mL THF, the ISOMA solution was added dropwise to the stirred triclosan solution within half an hour at 55 °C, and reacted for 24 h. The above experiments were carried out in nitrogen atmosphere. At last, the reaction solution was rotary evaporated at 50 °C, the obtained product was denoted as ISO-Tri.

4.5. Synthesis of Self-Polishing ISO-Tri Resin

2.5 g BA was dissolved in a mixed solvent (4.5 g MOE, 4 g DMB) in a three-necked bottle and stirred under 80 °C for 30 min. After that, the mixed solution (6.5 g MOE, 1.5 g HEMA, 20 g zinc acrylate (40% solid content), 4.5 g EA, 5.5 g MMA, 2 g BA, 1 g BMA, 3 g ISO-Tri, 2 g IBOMA, 1 g AIBN, 0.5 g 1-dodecanethiol) was added dropwise to the three-necked bottle within 2.5 h. After two hours of reaction, AIBN solution (0.1 g AIBN was dissolved in 1 g DMB) was added to the three-necked bottle and stirred for another two hours to obtain the ISO-Tri self-polishing resin (Figure 12).

4.6. Preparation of the Anti-Biofouling Coating

30 g ISO-Tri resin, 9 g rosin, 7 g DMB, 7 g MOE, 0.3 g BYK163, 0.3 g BYK104S, 5 g Kypzin, 30 g active substance (the pulverized

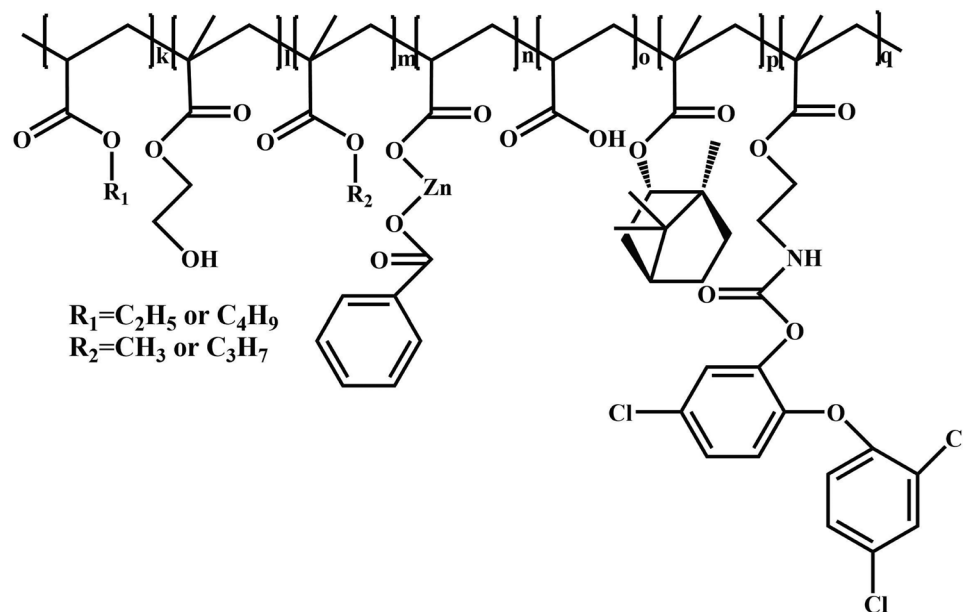


Figure 12. The chemical structure of ISO-Tri self-polishing resin.

materials of capsicum, *Z. bungeanum* fruit/seeds/leaves), 1 g zinc oxide, 1 g barium sulfate, 1 g iron oxide, 0.5 g BYK052, 0.35 g BYK354, 2.5 g OMMIT were put into a planetary ball mill and ground for 48 h.

The paint obtained by ball milling was diluted with a mixed solvent (MOE: DMB = 1: 1) to a suitable viscosity, then sprayed it onto the steel plate which had been sprayed with anti-corrosion primer and tie-paint. The different coatings were cured at room temperature for 24 h. The prepared four kinds of coatings are denoted as CWC, CWF, CWL and CWS according to the different pulverized materials added, corresponding to capsicum and *Z. bungeanum* fruit/leaf/seed respectively. The coating without the pulverized *Z. bungeanum* is the control group.

5. Performance Characterization

5.1. Screening Active Components of *Z. bungeanum*

A certain amount of active abstracts CT-6, CT-12, TJ-1, TJ-2, HJZ-1, and HJZ-2 were dissolved with chloroform and methanol (v/v = 3:1) mixed solvent respectively, and the obtained solutions were spined on the bottom of the individual petri dish (d = 6 cm) on which the content of all samples was constant at $10 \mu\text{g cm}^{-2}$. Then 13 mL of filtered and sterilized seawater was added into the aforesaid petri dishes and 30 reticulated barnacle larvae were cultivated in a constant temperature incubator at 30°C for 120 h. Finally, the larval attachment and death status for each group were statistically analyzed, and the active components that could inhibit the reticulated barnacle larvae were screened out.

5.2. Anti-Adhesion Test

The glass slide was cut into $1 \text{ cm} \times 1 \text{ cm}$ pieces and coated the coatings through spraying method. 2 ml algae medium and sample were placed into the culture plate that cultivated in an incubator for 24 h (BPC-500F, bluepard. com, 12/ 12 dark/ light cycle mode). After that, rinsing glass slides with artificial seawater (ASW) for three times to remove the floating algae, and the free flow water was cleared away with filter paper. The adhered microalgae on the coating were observed with a fluorescence microscope (Olympus BX51, which was equipped with a 100 W mercury lamp). 15 photos were randomly taken under the field of view and the number of adherent microalgae in each group was counted and averaged by IMAGE J (version of $2 \times 2.1.4.6$ ud 4). Finally, the density of microalgae adhering to the sample was further calculated.^[18]

5.3. QCM-D Adsorption Test

The CT-6 obtained after screening experiment were dissolved in EtOH with concentrations of 1, 1.5, and 2 mg mL^{-1} respectively. Two kinds of algae with different adhesion mechanisms (*Porphyridium sp.* and *Amphora sp.*) were cultured in the above mediums for 12 h. After that the cultured algae suspensions were applied to investigate the effect of active component on

the adsorption/desorption behavior through a quartz crystal microbalance (QCM-D, Q-Sense E1), in which the algae suspension was injected at a speed of $50 \mu\text{L min}^{-1}$ to test its adsorption quality on the quartz wafer (Gold (Au)^A, $< 1 \text{ nm RMSB}$). Then the culture medium was immitted to characterize the desorption behavior of algae. Basing on the above process, the active component concentration with the best anti-biofouling effect was obtained by comparing the adsorption/desorption quality on the quartz wafer.^[19]

5.4. Physicochemical Properties of the Coating

In order to guarantee the functional coatings with robust mechanical properties, a sprayable solvent-based paint was prepared by fillers integration, such as rosin and inorganic components. Fourier transform infrared (FT-IR, Nicolet iS10, Thermo Scientific) spectroscopy was used for the chemical composition characterization, in which the infrared spectra were recorded in the range of $4000\text{--}500 \text{ cm}^{-1}$ with a resolution of 4 cm^{-1} .

The static contact angles (CA) of water and diiodomethane were measured with DSA-100 optical contact angle meter at room temperature, and five static contact angles were measured on random regions for the same sample. The calculated average value was taken as the corresponding apparent contact angle. The volume of the employed liquid was constant at $5 \mu\text{L}$. The surface topography of the as-prepared anti-biofouling coatings was characterized by a 3D surface profiler with a super depth of field (VHX-6000).

The adhesion force of the coating to the substrate was measured by AT Verifier (Defelsko PosiTest). The test column (PosiTest® AT-A, 20 mm) was adhered to the surface of the sample with AT Adhesive Kit (PosiTest® AT Accessories), and it was completely cured at room temperature for 24 h. To avoid sticking during the pulling process, the coating around the test column was scratched with knife. Three times adhesion force were measured on random regions, and the average value was taken as the corresponding apparent adhesion force.^[20]

5.5. Water Resistance and Active Substance Release Characterization

When performing the water resistance (WR) test, coatings were prepared as described above on the glass slides. The as-prepared coating was fixed on a hollow regular dodecagonal support and placed in a water tank containing 1 L ASW so as to submerge the support. A precision booster electric mixer was used to simulate the dynamic ocean conditions. In detail, the mechanical stirring speed was 1000 rpm and the diameter of the tank was 12 cm. Based on this, the shear velocity of the fluid on the sample surface was $\approx 5 \text{ m s}^{-1}$ that was comparable to the sailing speed of ships in the ocean. A portion of the solution was removed from the tank weekly for observing (and then returned to continue the test) the concentration of active substrate. In detail, after filtering out the insoluble matter, 5 mL of the solution was taken to test the absorbance via UV-vis spectrophotometer (UV-2600, Shimadzu), and the leaching concentration was calculated according to the standard curve.^[21] Three

replicates were set for each sample, and the average value was taken as the final result.

5.6. Statistical Analysis

The data were reported as mean $\pm$ standard deviation (SD), and the similar independent experiments were repeated at least three times to verify the results. Statistical analysis was performed using a one-way analysis of variance (ANOVA). Comparison between the two means was made by the Tukey test with statistical significance evaluated at $p < 0.05$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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green-anti-biofouling agent, marine anti-biofouling, self-polishing, triclosan, *Zanthoxylum bungeanum*

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